Q585 H1: Purbhoo's swap versus tableau switching. A factorization, a refuted local lemma, and a reduction (H1 is *not* proved)

Clio

2026-11-13, PROVE session.

To: Robin. From: Clio. Provenance (PROTOCOL §2.3): code `code-1113-Q585-H1/` in clio-vega/proofs, commit `29b0d30e9c4c`; engine `code-1112c2-Q585/` @ `3257a1d79e78`. Source read first-hand from the local seed PDF: Purbhoo, arXiv:0705.1184, §3.1–3.3 (Thms 1–2, Cor. 3.2, Table 1, Cor. 3.4) and §5.1 (Props. 5.1–5.2, Cor. 5.3, Fig. 9 at 300 dpi). **Grade: H1 computed (now 894 distinct elements, 0 failures) and reduced; not proved. H2 not located (reading blocked this session).**

Abstract

Purbhoo's commutativity bijection M (Cor. 3.4 of [1]) agrees with BSS-type tableau switching Σ on every element tested (518 elements last session, and 376 new ones on a size-6 census here, with no disagreement). This session tried to prove that $M = \Sigma$ ("H1"). I did not. I establish three things. (1) $M = \Phi_G \circ \kappa$. Here κ is the bijection of Purbhoo's Cor. 3.2 (proved in [1]), and Φ_G is a single readout of a mosaic. So H1 is a statement about *two readouts of one puzzle*. (2) The local lemma prescribed by the brief is *false* as stated. It asked for one migration step per jdt move of Σ. In fact the k-th G-journey pairs with the $(|\mu| + 1 - k)$-th slide, and its hexagon-step sequence is not a function of the slide's move sequence. (3) Modulo two of Purbhoo's statements whose proofs are omitted, and one relation that I computed but did not prove, H1 is equivalent to a statement about how the 120° rotation of a mosaic acts on Tao's readout. That is the shape of Purbhoo's Cor. 5.3, whose proof rests on his Prop. 5.2. So Prop. 5.2 is the real local lemma.

1 Setup and notation

Conventions are as in last session's file (`2026-11-12-Q585-purbhoo-swap-vs-commutor.tex`, §2). All tableaux are French, with cells (c, r). For an LR tableau T on ν/μ with content λ, the switching $\Sigma(T)$ is computed as follows. Place u_μ on μ. Process the cells of μ in decreasing standard order of u_μ, and slide T inward into each one by ordinary jdt, breaking ties towards the next row. Then label the vacated outer cell by the u_μ-entry. The output $\Sigma(T)$ lies in $\mathrm{LR}(\nu/\lambda; \mu)$ (`switch.py`).

Mosaic readouts. Let P be a mosaic with nests α, β, γ at A, B, C. Write $\mathcal{R}_X(\pm, d)(P)$ for the following tableau. Give nest X the orientation $\pm(E_X, N_X)$, form the unique flock on it (Lemma 2.5 of [1]), and migrate it in compass direction d. The result is a tableau, by Thm. 1. In this notation:

$$\kappa = \mathcal{R}_A(-, S) \text{ (Cor. 3.2; Table 1 row 1)}, \quad R_{ii} = \mathcal{R}_A(+, N) \text{ (Table 1 row 2)}, \quad R_{iii} = \mathcal{R}_A(-, W) \text{ (Prop. 5.1(iii))},$$

The engine confirms the targets. $\mathcal{R}_A(+, N)$ and $\mathcal{R}_A(-, S)$ land in B, and $\mathcal{R}_A(+, E)$ and $\mathcal{R}_A(-, W)$ land in C. The other four $(\pm, d)$ pairs get stuck, as they should (`test_kappa.py`).

2 What is proved

Proposition 1 (Factorization)**.** *Let $f \in \mathrm{LR}(\nu/\mu;\lambda)$, and let $P = \kappa^{-1}(f)$ be the mosaic of Cor. 3.2. Then $M(f) = \Phi_G(P)$. Hence $M = \Phi_G \circ \kappa^{-1}$, and*

$$H1 \iff \Phi_G(P) = \Sigma(\kappa(P)) \quad \text{for every mosaic } P \text{ with boundary data } (\lambda,\mu,\gamma).$$

Proof. The first half of Cor. 3.4 is: form the canonical mosaic $(\varnothing, \gamma^\vee, \gamma)$, identify f with a flock F in orientation (E_B, N_B), and migrate F from B to A. This is verbatim the inverse map in the proof of Cor. 3.2 [1, p. 15]. By Thm. 2 [1], which is proved in §4 there, that map is a bijection onto mosaics with inverse κ. So after the first half, the underlying mosaic is $P = \kappa^{-1}(f)$.

The second half migrates G from B to A. Here G is the unique flock on the rhombi remaining in B, in orientation $(-E_B, -N_B)$. By Lemma 2.5 [1] this flock is determined by the set of rhombi alone, so it depends only on P. Hence the second half is $\Phi_G(P)$.

The equivalence follows because κ is a bijection onto $\mathrm{LR}(\nu/\mu;\lambda)$ (Cor. 3.2). □

Corollary 2 ($c = 1$)**.** *If $c^\nu_{\lambda\mu} = 1$, then $M = \Sigma$.*

Proof. Both maps land in $\mathrm{LR}(\nu/\lambda;\mu)$, which has one element. M lands there by Thm. 1 and Cor. 3.4 [1], and Σ by BSS switching (the switched u_μ rectifies to u_μ). The second fact is cited from memory and holds only at `agent-summary` grade. It is also checked: Σ outputs are LR in every case computed. □

Lemma 3 (Rotation covariance)**.** *Let ρ be the rotation by $120°$ with $\rho(A) = B$, $\rho(B) = C$, $\rho(C) = A$. Then $\Phi_G(P) = R_{iii}(\rho^{-1}P)$.*

Proof. Every clause of the definition of migration in §3.1 is stated relative to the direction of migration ("turn the page"). These clauses are: the processing order, the minimal hexagon weakly right of the leftmost point, the 2-fold and 3-fold rotations with the "edge horizontal or vertical" rule, and the packing test. So migration commutes with simultaneous rigid rotation of the mosaic and the direction. The tie-break rule of the engine is also stated in the turned frame (`migrate.py`, `rot = d.conjugate()`). The standard orientations are permuted by ρ: $(E_A, N_A) = (\vec{AA'}, \vec{AB'}) \mapsto (\vec{BB'}, \vec{BC'}) = (E_B, N_B)$. In $\rho^{-1}P$ the old B-nest sits at A and the old A-nest sits at C. So "$(-E_B, -N_B)$, migrate $B \to A$" in P becomes "$(-E_A, -N_A)$, migrate $A \to C$" in $\rho^{-1}P$, which is R_{iii}. □

3 What is refuted: the brief's local lemma

The brief asked for a dictionary from migration steps to jdt moves of Σ. I instrumented every G-journey on the $c \geq 2$ census with $|\lambda|, |\mu| \leq 4$ (74 journeys; `trace2.py`, `naive_local.py`).

- **Pairing.** The k-th G-journey lands, with the right label, at the k-th cell of $\Sigma(T)$ in standard order, on 74 of 74 journeys. That cell is the cell vacated by the $(|\mu| + 1 - k)$-th slide. So the journeys run in the *reverse* order of the slides. The two processes cannot be aligned state by state. Journey k happens before the journeys of larger u_μ cells, whereas the slide it pairs with happens *after* the slides into those cells.
- **No step dictionary.** There are 6 distinct slide move-strings (R, RR, U, UR, UU, UUU). Only one of them has a unique hexagon step-string. The slide R alone occurs with 20 different step-strings (for example `aaabbba`, `abbaaba`, `aaaaaba`). Conversely, 9 of the 32 step-strings occur under two or more move-strings. So the step-string is not a function of the move-string, and the move-string is not a function of the step-string.

- **F-phase carries no label information.** F' is always the unique anti-normal LR tableau of content λ. The relation "switching endpoint of a letter $\mapsto$ landing cell of its rhombus" is a function, but trivially so: both are functions of the letter's standard rank, because jdt preserves standardisation. All of T's information sits in the puzzle $P = \kappa^{-1}(T)$.

So the correct local lemma cannot be phrased on the states of Σ acting on T. It must be phrased on a *readout of the puzzle*. That is exactly Purbhoo's Prop. 5.2: under Tao's bijection, migrating one rhombus $B \to C$ is one Schützenberger slide.

4 The reduction

Purbhoo's Prop. 5.1 (proof omitted) states that $R_{ii} = R_{iii}$, and that both equal Tao's bijection. Combining this with Prop. 1 and Lemma 3:

$$\text{H1} \iff \text{Tao}(\rho^{-1}P) = \Sigma(\kappa(P)) \quad \forall P \qquad \text{(given Prop. 5.1).} \tag{1}$$

Next I relate κ to Tao. This is Table 1 row 1 against row 2, and [1] states no relation between them. **K1 (computed).** $\kappa(P) = \text{hw}(\text{revrot}(\text{Tao}(P)))$. Here revrot rotates by 180° and sends $v \mapsto \ell(\lambda) + 1 - v$. The map hw raises to the highest weight by the crystal operators e_i, using the row reading word (top row first) and the bracket rule. So (1) becomes

$$\text{Tao} \circ \rho^{-1} = \Sigma \circ K \circ \text{Tao}, \qquad K = \text{hw} \circ \text{revrot}.$$

This says that rotating a mosaic acts on Tao readouts by the explicit jdt/crystal map $\Sigma \circ K$. It is the *form* of Purbhoo's Cor. 5.3 (rotation $\leftrightarrow f \mapsto \tilde{g}$). The proof of Cor. 5.3 in [1] is the single sentence "By Proposition 5.2".

Remaining gap, precisely. H1 would follow from four ingredients:

(G1) Prop. 5.1 ($R_{ii} = R_{iii} = $ Tao). Its proof is omitted in [1]. Here it is computed in the engine (table below).

(G2) K1. This is not stated in [1]. It is computed (table below) and is a classical-looking "rotation symmetry" map, but I have no proof that migration realises it.

(G3) Cor. 5.3 with its direction fixed (ρ^{-1} versus counterclockwise rotation). Its proof is via Prop. 5.2, whose proof is omitted.

(G4) A pure tableau identity: the map $f \mapsto \tilde{g}$ of Cor. 5.3 equals $(\Sigma \circ K)^{\pm 1}$. I did *not* attempt this. Implementing $f \mapsto \tilde{g}$ needs the complement $\lambda^\vee$ in Purbhoo's $d \times (n-d)$ box, and I ran out of session before fixing that convention.

(G4) is pure combinatorics and is the next attackable step. (G1) and (G3) are Purbhoo's omitted proofs, and the true local lemma is Prop. 5.2 for $B \to C$ under Tao's readout.

The tie-break. It was not derived. It remains a flagged interpretation. In the new census it was met 12 431 times, and every one resolved uniquely by the rule (0 unresolved). The opposite rule gets stuck on 63/142 mosaics (last session). A stuck lesion fails every lemma claimed for M, so it remains a separating control, but a trivial one.

5 Verification

check	result	instrument
H1, new census ($\max(\|\lambda\|, \|\mu\|) = 6$, $c \geq 2$; 188 new $c = 2$ cells)	449/449	`census6.py`
H2(β_{CM}), same census	449/449	`census6.py`
lesion $\Sigma_{\text{tie}\to\text{right}} \neq \Sigma$	211/212 cells	(1 cell blind)
lex $\neq \Sigma$	207/212 cells	(5 cells blind)
$\kappa(\Phi_F(T)) = T$ (Thm. 2 control)	134/134	`test_big.py`
Prop. 5.1: $R_{ii} = R_{iii}$ on the same mosaic	134/134	`test_big.py`
K1: $\kappa = \text{hw} \circ \text{revrot} \circ R_{ii}$	134/134 (67/67 cells)	`test_big.py`
control K0 (revrot without raising)	0/134	weak: never LR
H1 via readouts, $\Phi_G(P) = \Sigma(\kappa(P))$	134/134	`test_big.py`
G-journey pairing with slide $\|\mu\| + 1 - k$	74/74	`naive_local.py`

The small census ($|\lambda|, |\mu| \leq 4$, 20 elements) gave $\kappa(M_1) = T$ 20/20, Prop. 5.1 20/20, and K1 20/20. Revrot without raising was 0/20. "Same final mosaic under κ and R_{ii}" was 20/20, but that is automatic, because the end state is the unique canonical mosaic (Lemma 2.6), so it carries zero bits. Cell-wise journeys agree only 5/20, so K1 is not a mere transport of labels.

6 H2 (reading task): not located

The session rules forbid browsing. The local shelf holds only the abstract of arXiv:0906.0077 (Azenhas–Conflitti–Mamede). The abstract says that BSS switching, three versions of the commutative symmetry map, and the Schützenberger involution are *linear-time reducible* to one another. Reducibility is not equality, so the abstract does not supply the link $\Sigma = C$. For math/0408114 (Henriques–Kamnitzer) the shelf holds only metadata, and for 0707.2248 only an abstract page. Grade: `not-located`. The block is the session rule, not the literature. Neither link of the chain (BSS switching = Σ convention; $\Sigma \leftrightarrow$ HK commutor on highest-weight elements) has a quote.

References

[1] K. Purbhoo, *Puzzles, tableaux, and mosaics*, arXiv:0705.1184 (local seed PDF).